
A Median Perspective on Unlabeled Data for Out-of-Distribution Detection

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Abstract

1 Out-of-distribution (OOD) detection is a deployment-critical requirement for machine learning systems, yet labeled OOD data is rarely available when models
2 enter open-world environments. This paper makes three contributions toward using
3 unlabeled in-the-wild data for OOD detection. First, we introduce Medix, a
4 median-centric framework that extracts candidate outliers from unlabeled mixtures
5 of in-distribution (InD) and OOD samples and then trains an OOD detector with the
6 extracted candidates and labeled InD data. Second, we show why the median is a
7 useful filtering primitive in this setting: its robustness to contamination yields error
8 bounds for Medix that control both InD samples incorrectly flagged as outliers and
9 OOD samples incorrectly retained as inliers. Third, we validate this contribution
10 empirically across open-world benchmarks, where Medix outperforms existing
11 methods and supports the theoretical finding that median-based filtering can recover
12 useful OOD candidates from mixed wild data.
13

14 1 Introduction

15 Machine learning systems deployed in real-world environments must often decide when an input
16 lies outside the training distribution. This OOD detection problem is central to safety and reliability
17 because samples from unknown categories should not be forced into known labels. Standard neural
18 networks, however, are vulnerable to such inputs and typically lack a built-in mechanism for detecting
19 or handling OOD data in practice (Nguyen et al., 2015).

20 The core difficulty is that unknown distributions are absent during supervised training, making it
21 hard for a model to distinguish OOD inputs from in-distribution (InD) data at inference time. A
22 promising direction is to use additional unlabeled “in-the-wild” data collected from the model’s
23 operating environment (Katz-Samuels et al., 2022a; Du et al., 2024a). For example, Katz-Samuels
24 et al. (2022a) use unlabeled wild data to regularize training while preserving the primary goal of
25 classifying labeled InD samples. This line of work is attractive because wild data can be collected
26 after deployment and may better reflect the OOD distribution encountered at test time.

27 The challenge, and the starting point for our contribution, is that wild data is not a clean OOD
28 dataset. It is a mixture of InD and OOD samples, with no label indicating which component generated
29 each point. Consequently, a detector trained directly on this mixture lacks a reliable set of negative
30 examples. The in-the-wild setting therefore remains underexplored, especially from a theoretical
31 perspective: to the best of our knowledge, Du et al. (2024a) is the only prior work that provides a
32 formal foundation for this setting.

33 Motivated by recent evidence that median-based procedures are effective for data pruning (Acharya
34 et al., 2024), we ask whether the same robustness principle can be turned into an OOD filtering
35 mechanism. The central question of the paper is:

How can median-based methods leverage unlabeled wild data to facilitate OOD detection with theoretical guarantees?

We answer this question with a median-centric framework for OOD detection. Our method, Medix, formulates outlier extraction as a median-based filtering problem over gradients and then uses the extracted candidates to train a detector. This contribution is both algorithmic and theoretical: we bound the fraction of InD samples incorrectly flagged as outliers and show that the bound is controlled by a contamination effect, which remains manageable when the OOD proportion is below 50%, and a concentration effect, which uses sub-Gaussian InD gradient behavior to limit deviations from the InD mean gradient. These guarantees make the filtering role of the median explicit and show why Medix can remain reliable even when the wild dataset is a contaminated mixture.

Empirically, we evaluate the same contribution from two reviewer-relevant angles: performance against methods trained only on InD data and performance against methods that also use auxiliary unlabeled data. On CIFAR-100 (Krizhevsky et al., 2009), Medix improves over the strong KNN+ baseline (Sun et al., 2022) by an average of 40.98% in FPR95. Unlike Outlier Exposure (Hendrycks et al., 2019), which assumes clean auxiliary outlier data that is separable from InD data, Medix operates on mixed wild data. Compared with WOODS (Katz-Samuels et al., 2022a), Medix reduces average FPR95 by 1.32% on CIFAR-100 and 2.60% on CIFAR-10.

Our key contributions are:

- C1)** We propose Medix, a median-centric greedy framework for unlabeled wild data that first filters candidate outliers and then trains an OOD detector with those candidates and labeled InD samples. The primary methodological contribution is the filtering stage.
- C2)** We establish theoretical guarantees for median-based filtering in unlabeled mixtures. The bounds control both InD samples flagged as outliers and OOD samples retained as inliers through contamination and concentration effects, with the contamination term remaining controlled when the OOD proportion is below 50%.
- C3)** We evaluate Medix across eleven InD-OD pairs against 20 competitive baselines. Medix outperforms all baselines in these comparisons, and a controlled visualization shows a low outlier-extraction error rate (e.g., 12.5%; see Figure 2), supporting the filtering claims behind C1 and C2.

2 Preliminaries and Problem Setup

This section fixes the problem setting for the contributions above. We define the labeled InD data, the unlabeled wild mixture that Medix filters, the models used for classification and OOD detection, and the evaluation objective used throughout the theory and experiments.

Labeled In-Distribution Data. We begin with the supervised data that anchors both the reference gradient and the final detector. Let $\mathcal{X}$ be the input space and $\mathcal{Y} = \{1, \dots, K\}$ be the InD label space. A labeled dataset $\mathcal{S}_{\text{in}} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ is generated by sampling n pairs independently and identically distributed (i.i.d.) from $\mathbb{P}_{\mathcal{X}\mathcal{Y}}$, an unknown joint probability distribution over $\mathcal{X} \times \mathcal{Y}$. We denote the marginal distribution of $\mathbb{P}_{\mathcal{X}\mathcal{Y}}$ on $\mathcal{X}$ by $\mathbb{P}_{\text{in}}$. This labeled set $\mathcal{S}_{\text{in}}$ is used to train the InD model and to define the InD reference against which wild samples are filtered.

Out-of-distribution detection. We study the deployment setting in which a model trained on labeled InD data later encounters inputs from classes not represented during training, i.e., for some label $y \notin \mathcal{Y}$. The desired behavior is not to assign such inputs to one of the known classes, but to identify them as OOD. Thus the inference-time goal is a binary decision: determine whether an input is drawn from the InD distribution or from an OOD source.

Unlabeled wild data. The main resource exploited by Medix is unlabeled wild data rather than labeled OOD data. Since the OOD sample space can be large and costly to annotate, we introduce $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_m\}$ into the learning framework, following the real-world motivation of Katz-Samuels et al. (2022a). This wild set is a deployment-collected mixture of InD and OOD samples, and the contribution of Medix is to extract useful candidate outliers from this mixture. Similar to Du et al. (2024a); Katz-Samuels et al. (2022a), we characterize its marginal distribution with the Huber

84 contamination model (Huber, 1964)

$$\mathbb{P}_{\text{wild}} := (1 - \pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}, \quad \pi \in (0, 1], \quad (1)$$

85 where π denotes the contamination proportion and $\mathbb{P}_{\text{out}}$ denotes the OOD distribution over $\mathcal{X}$. The
86 case $\pi = 0$ contains no OOD samples and is therefore trivial for the filtering problem considered
87 here.

88 **Models and Loss Functions.** We use two models because the paper contributes both a way to exploit
89 InD supervision and a way to train an OOD detector from filtered wild data. Let $f_\phi : \mathcal{X} \rightarrow \mathbb{R}^K$ be
90 the InD classifier parameterized by $\phi \in \Phi$, with outputs over the $K = |\mathcal{Y}|$ InD classes and supervised
91 loss $\ell : \mathbb{R}^K \times \mathcal{Y} \rightarrow \mathbb{R}$. For OOD detection, we introduce a separate binary classifier $g_\theta : \mathcal{X} \rightarrow \mathbb{R}$,
92 parameterized by $\theta \in \Theta$. Its binary loss is denoted by $\ell_b(g_\theta(x), y_b)$, where $y_b \in \mathcal{Y}_b := \{y_+, y_-\}$,
93 with $y_+ > 0$ representing InD and $y_- < 0$ representing OOD.

94 **Learning objective.** The learning objective reflects the paper’s central structure: use $\mathbb{P}_{\text{in}}$ to learn a
95 reliable InD classifier f_ϕ , use $\mathbb{P}_{\text{wild}}$ to obtain candidate outliers, and train g_θ to separate InD from
96 OOD inputs. At test time, we evaluate the resulting system with the following metrics:

$$\begin{aligned} \downarrow \text{FPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{out}}^{\text{test}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{TPR}(g_\theta) &= \mathbb{E}_{\mathbf{x} \sim \mathbb{P}_{\text{in}}} [\mathbb{I}\{g_\theta(\mathbf{x}) = \text{in}\}], \\ \uparrow \text{Acc}(f_\phi) &= \mathbb{E}_{(\mathbf{x}, y) \sim \mathbb{P}_{\mathcal{X} \times \mathcal{Y}}} [\mathbb{I}\{f_\phi(\mathbf{x}) = y\}], \end{aligned}$$

97 where $\mathbb{I}\{\cdot\}$ denotes the indicator function, and $\mathbb{P}_{\text{out}}^{\text{test}}$ represents the OOD test distribution.

98 3 Method: Median-Centric Framework for OOD Detection

99 This section presents the main methodological contribution: **Medix**, a median-centric framework for
100 turning unlabeled wild data into useful OOD supervision. The framework is deliberately organized
101 into two stages so that the filtering contribution is explicit. **1) Outlier Extraction:** Medix isolates
102 candidate OOD samples from the unlabeled wild mixture (Section 3.1). **2) Detector Training:** it
103 trains a binary OOD detector using labeled InD data and the extracted candidates (Section 3.2); for
104 this stage, we follow the protocol introduced by Du et al. (2024a). The rest of the section explains
105 how the median-based extraction stage makes the use of mixed wild data practical in open-world
106 environments.

107 3.1 Extracting Candidate Outliers from the Wild Data

108 To isolate potential outliers from $\mathcal{S}_{\text{wild}}$, Medix uses gradients as the representation on which the
109 median filter operates. The gradients come from a classifier f_ϕ trained only on $\mathcal{S}_{\text{in}}$, so the filtering
110 score is anchored in labeled InD behavior rather than in any labeled OOD examples. We now define
111 this extraction stage.

112 **Reference gradient estimation from InD data.** The first ingredient in Medix is an InD reference
113 gradient. We estimate it by training f_ϕ on $\mathcal{S}_{\text{in}}$ through empirical risk minimization (ERM):

$$\phi_{\mathcal{S}_{\text{in}}} \in \arg \min_{\phi \in \Phi} \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi), \quad \text{where } \mathcal{L}_{\mathcal{S}_{\text{in}}}(f_\phi) = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \ell(f_\phi(\mathbf{x}_i), y_i), \quad (2)$$

114 where $\phi_{\mathcal{S}_{\text{in}}}$ denotes the learned parameters. After training, we compute the mean InD gradient $\bar{\nabla}_{\text{in}}$,
115 which becomes the reference point for measuring whether wild samples behave like InD data:

$$\bar{\nabla}_{\text{in}} = \frac{1}{n} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{S}_{\text{in}}} \nabla \ell(f_{\phi_{\mathcal{S}_{\text{in}}}}(\mathbf{x}_i), y_i). \quad (3)$$

116 Thus $\bar{\nabla}_{\text{in}}$ is the InD anchor used by the median filter to quantify deviations in the wild mixture.

117 **Motivation.** The key empirical observation behind C1 is that adding OOD samples to the wild
118 set should move the median gradient away from the InD reference. We test this using CIFAR-10
119 (Krizhevsky et al., 2009) as InD data and SVHN (Netzer et al., 2011) as OOD data. The initial $\mathcal{S}_{\text{wild}}$
120 contains 10,000 CIFAR-10 samples disjoint from the training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$ and compute

121 $\bar{\nabla}_{\text{in}}$. We then add SVHN samples incrementally and track the L_2 deviation between $\bar{\nabla}_{\text{in}}$ and the
 122 element-wise median (EWM) of wild-data gradients:

$$\left\| \bar{\nabla}_{\text{in}} - \text{EWM} \left(\left\{ \nabla \ell \left(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in S_{\text{wild}}} \right) \right\|.$$

123 Figure 1 shows a clear monotonic increase in this deviation as OOD samples are added. This motivates
 124 the Medix filter: if removing a sample substantially moves the EWM closer to the InD reference, that
 125 sample is a strong OOD candidate. The same observation also motivates the stopping rule, which
 126 terminates filtering when the change in L_2 deviation between consecutive iterations falls below ϵ .

127 **Filtering potential outliers from unlabeled wild data.** Motivated by Figure 1, we express the
 128 filtering contribution as an optimization problem: identify the subset of wild samples whose EWM
 129 gradient is closest to the InD reference, and treat the complement as candidate outliers.

$$\mathcal{S}_{\text{in}}^* = \arg \min_{\mathcal{S} \subseteq \mathcal{S}_{\text{wild}}} \left\| \bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}}) \right\|, \text{ where } G_{\mathcal{S}} = \left\{ \nabla \ell \left(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i} \right) \right\}_{i \in \mathcal{S}}. \quad (4)$$

130 Here $\text{EWM}(\cdot)$ denotes the element-wise median function, and $\hat{y}_{\tilde{\mathbf{x}}_i}$ represents the predicted
 131 label for a wild sample $\tilde{\mathbf{x}}_i$. For notational simplicity, we define $G_{\mathcal{S}} = \left\{ \nabla \ell(\tilde{\mathbf{x}}_i) \right\}_{i \in \mathcal{S}}$.
 132 This objective makes the intended contribution explicit: the selected subset $\mathcal{S}_{\text{in}}^*$ is the portion of
 133 $\mathcal{S}_{\text{wild}}$ whose median gradient most resembles InD behavior. Guided by Figure 1, we interpret $\mathcal{S}_{\text{in}}^*$
 134 as the likely InD portion of the wild set and use
 135 $\mathcal{S}_{\text{out}}^* = \mathcal{S}_{\text{wild}} \setminus \mathcal{S}_{\text{in}}^*$ as the corresponding OOD candidate set.
 136
 137
 138

139 Solving (4) exactly can be computationally pro-
 140 hibitive for large $\mathcal{S}_{\text{wild}}$. Medix therefore imple-
 141 ments the filtering contribution with a greedy leave-
 142 one-out approximation, summarized in Algorithm 1.
 143 The algorithm repeatedly removes samples whose
 144 exclusion most reduces the median-gradient devia-
 145 tion from the InD reference.

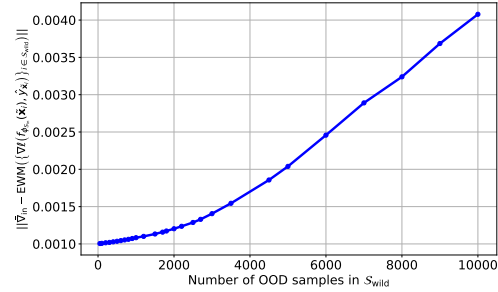


Figure 1: Median-gradient deviation increases as more OOD samples are added to $\mathcal{S}_{\text{wild}}$, motivating the Medix filtering criterion.

146 At iteration t , Medix computes the EWM of the remaining wild gradients and compares it to $\bar{\nabla}_{\text{in}}$. We
 147 denote this distance by $d_t = \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$, where $\mathcal{S}$ is the current remaining wild set. For
 148 each sample $i \in \mathcal{S}$, the algorithm recomputes the EWM after removing that sample, $\text{EWM}(G_{\mathcal{S} \setminus \{i\}})$,
 149 and measures the resulting distance to $\bar{\nabla}_{\text{in}}$. The score $\delta_i = d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$ captures
 150 how much closer the median gradient moves toward the InD reference when i is removed. Medix
 151 selects the k samples with the largest drops as candidate OOD samples and repeats until δ_i no longer
 152 changes substantially or the maximum iteration count is reached. The threshold ϵ operationalizes the
 153 trend in Figure 1: once enough OOD samples have been removed, the remaining set is already close
 154 to the InD reference, and continuing to filter risks removing InD samples.

Algorithm 1 Iterative Outlier Detection via Medix

Require: $\bar{\nabla}_{\text{in}}$, $G_i = \nabla \ell(f_{\phi_{S_{\text{in}}}}(\tilde{\mathbf{x}}_i), \hat{y}_{\tilde{\mathbf{x}}_i})$, maximum iterations T , hyperparameters ϵ , k

- 1: Initialize $\mathcal{S} \leftarrow \mathcal{S}_{\text{wild}}$ (wild set), $\mathcal{O} \leftarrow \emptyset$ (outliers), $t \leftarrow 0$ (iteration), $d_t \leftarrow 0$, $\delta_{\text{max}} \leftarrow \infty$ (deviations), $\mathcal{I}_k \leftarrow \emptyset$
- 2: **while** $t \leq T$ or $|\delta_{\text{max}}| > \epsilon$ **do**
- 3: $\mathcal{O} \leftarrow \mathcal{O} \cup \{\tilde{\mathbf{x}}_i : i \in \mathcal{I}_k\}$ ▷ Add outliers to set $\mathcal{O}$
- 4: $d_t \leftarrow \|\text{EWM}(G_{\mathcal{S}}) - \bar{\nabla}_{\text{in}}\|$ ▷ Compute L2 deviation
- 5: **for each** $i \in \mathcal{S}$ **do**
- 6: $\delta_i \leftarrow d_t - \|\text{EWM}(G_{\mathcal{S} \setminus \{i\}}) - \bar{\nabla}_{\text{in}}\|$
- 7: **end for**
- 8: $\mathcal{I}_k \leftarrow \text{indices}(\text{top-}k(\{\delta_i\}_{i \in \mathcal{S}}))$ ▷ Select top- k indices
- 9: $\mathcal{S} \leftarrow \mathcal{S} \setminus \mathcal{I}_k$ ▷ Remove outliers from $\mathcal{S}$
- 10: $\delta_{\text{max}} \leftarrow \max_{i \in \mathcal{S}} \{\delta_i\}$, $t \leftarrow t + 1$
- 11: **end while**
- 12: **return** $\mathcal{S}_{\text{out}} = \mathcal{O}$ ▷ Return detected outliers

3.2 Training the OOD Detector with Candidate Outliers

The second stage uses the filtered set $\hat{\mathcal{S}}_{\text{out}}$ as supervision for a binary OOD detector. Following Du et al. (2024a), we train g_θ to separate labeled InD samples from the candidate outliers produced by Medix. The decision boundary is thresholded at 0: labeled InD samples $\mathbf{x} \in \mathcal{S}_{\text{in}}$ should receive positive outputs, while candidate outliers $\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}$ should receive negative outputs. The objective is:

$$\begin{aligned}\mathcal{L}_{\mathcal{S}_{\text{in}}, \hat{\mathcal{S}}_{\text{out}}}(g_\theta) &= \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) + \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta), \\ \mathcal{L}_{\mathcal{S}_{\text{in}}}^+(g_\theta) &= \mathbb{E}_{\mathbf{x} \in \mathcal{S}_{\text{in}}} \mathbb{I}\{g_\theta(\mathbf{x}) \leq 0\}, \\ \mathcal{L}_{\hat{\mathcal{S}}_{\text{out}}}^-(g_\theta) &= \mathbb{E}_{\tilde{\mathbf{x}} \in \hat{\mathcal{S}}_{\text{out}}} \mathbb{I}\{g_\theta(\tilde{\mathbf{x}}) > 0\}.\end{aligned}\tag{5}$$

Because the 0/1 loss is non-differentiable, we optimize a differentiable sigmoid-based binary loss as a smooth surrogate. We also retain the InD risk in Equation 2 so that the detector-training stage does not sacrifice InD predictive accuracy. In this way, the extracted candidates from C1 become usable training signal for g_θ , and the experiments below evaluate whether this signal improves detection for OOD samples drawn from $\mathbb{P}_{\text{out}}$.

4 Theoretical Analysis

We now substantiate the main filtering contribution with theory. The goal is to show that the EWM stage in Medix controls both directions of filtering error: labeling true InD samples as outliers and retaining true OOD samples as inliers. The following theorems give high-probability upper bounds for these two error types; detailed proofs appear in Appendix C.

Theorem 4.1 (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$, and fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the EWM filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

Theorem 4.1 provides the first theoretical component of C2: it bounds how often Medix incorrectly removes InD samples. The bound separates the error into a *concentration term*, which decreases with more InD samples and captures finite-sample fluctuations in InD gradients, and a *contamination term*, which captures the influence of OOD gradients on the EWM. The key implication is that the median-based filter remains controlled when $\pi < 0.5$, matching the intuition that a median is robust to minority contamination. We next bound the complementary error: true outliers that are not flagged and remain in the retained set.

Theorem 4.2 (Outlier Misclassification Bound). *Assume that the gradients of OOD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and each coordinate is sub-Gaussian with variance proxy σ_{out}^2 . Suppose the mean OOD gradient μ_{out} satisfies a separation condition:*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta\sqrt{d},$$

for some $\Delta > 0$. Then, for any tolerance $\epsilon \in (0, \Delta)$ and confidence level $\delta \in (0, 1)$, with probability at least $1 - \delta$, the outlier misclassification rate satisfies:

$$\text{ERR}_{\text{out}} \leq \underbrace{2d \exp\left(-\frac{(\Delta - \epsilon)^2}{2\sigma_{\text{out}}^2}\right)}_{\text{Separation term}} + \underbrace{\sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}}_{\text{Concentration term}} + \underbrace{\frac{1 - \pi}{2\pi}}_{\text{Contamination term}}.$$

Theorem 4.2 completes the two-sided guarantee by bounding the fraction of OOD points mistakenly retained as InD. Together, Theorems 4.1 and 4.2 make the theoretical contribution explicit: Medix controls both removal errors and retention errors through interpretable contamination, concentration, and separation effects.

181 **Contamination Effect.** This term captures the cost of learning from a mixed wild set. In Theorem 4.1,
 182 the $\pi/[2(1 - \pi)]$ term increases with the OOD fraction and represents the risk that OOD gradients
 183 skew the EWM, while still remaining controlled for $\pi < 0.5$. In Theorem 4.2, the symmetric penalty
 184 $\frac{1-\pi}{2\pi}$ captures the difficulty of isolating OOD points when they are underrepresented.

185 **Concentration Effect.** This term explains why the gradient median is statistically stable. For InD
 186 data, sub-Gaussian coordinates keep gradients concentrated near $\bar{\nabla}_{\text{in}}$, which prevents the EWM from
 187 drifting sharply in finite samples. For OOD data, concentration around μ_{out} quantifies the chance
 188 that an OOD sample appears close enough to InD behavior to pass the filter. These effects decay as
 189 $1/\sqrt{m}$, where m is the number of samples from each class.

190 **Separation Effect.** This effect appears in Theorem 4.2 and quantifies when candidate outlier
 191 extraction is possible. The exponential term $\exp(-(\Delta - \varepsilon)^2/2\sigma_{\text{out}}^2)$ decreases as the OOD mean
 192 gradient moves farther from the InD mean gradient, showing that better separation makes it less likely
 193 for OOD gradients to fool the median filter.

Theorems 4.1 and 4.2 jointly establish the theoretical contribution of Medix: with high probability, median filtering robustly separates InD and OOD samples in unlabeled mixtures. InD removal error is bounded by contamination and concentration effects, while OOD retention error is controlled by separation, concentration, and reverse contamination effects.

194
 195 *Remark 4.3. Sub-Gaussian Assumption for Gradient Coordinates.* The concentration part of our
 196 guarantee uses the assumption that InD gradient coordinates are sub-Gaussian. We include empirical
 197 support for this assumption in Figure 4a: the InD gradient histogram is bell-shaped, concentrated
 198 around the mean, and light-tailed, which is consistent with sub-Gaussian behavior. Figure 4b further
 199 compares empirical gradient quantiles with Gaussian quantiles in a Q-Q plot (Wilk and Gnanadesikan,
 200 1968); the close alignment with the 45-degree reference line suggests that the empirical distribution
 201 resembles a Gaussian distribution, and Gaussian variables are sub-Gaussian. *To clarify the scope of*
 202 *the contribution, we also provide a looser guarantee that **removes the sub-Gaussian assumption** and*
 203 *requires only bounded second moments. This version is presented in Theorem C.3 (Appendix C.3).*
 204 *The rates degrade, but the core robustness claim for Medix persists under this broader assumption.*

205 5 Experiments

206 The experiments test the empirical side of the paper’s contribution structure. We ask whether the
 207 Medix filtering stage yields better OOD detectors across InD-ODD pairs, whether the gains hold
 208 against methods trained with and without wild data, and whether the extracted candidates align with
 209 the theory. We compare against 20 widely used baselines, run all experiments on hardware equipped
 210 with NVIDIA A100-SXM4-80GB GPUs, and will release code upon acceptance.

211 5.1 Models, Datasets, and Baselines

212 **Datasets.** We follow the protocol of Katz-Samuels et al. (2022a), which introduced learning OOD
 213 detectors with wild data, so that Medix is evaluated in the same setting as prior work. CIFAR-10 and
 214 CIFAR-100 serve as InD datasets ($\mathbb{P}_{\text{in}}$). For OOD testing, we use natural image datasets including
 215 PLACES365 (Zhou et al., 2017), SVHN, TEXTURES (Cimpoi et al., 2014), and LSUN-RESIZE &
 216 LSUN-C (Yu et al., 2015) as OOD datasets ($\mathbb{P}_{\text{out}}$). To evaluate the wild-data contribution directly,
 217 we construct $\mathbb{P}_{\text{wild}}$ by mixing a subset of InD data ($\mathbb{P}_{\text{in}}$) with OOD data ($\mathbb{P}_{\text{out}}$) using the default
 218 contamination parameter $\pi = 0.5$. For instance, when PLACES365 is the OOD test set, the wild
 219 mixture combines CIFAR and PLACES365, and PLACES365 is also used as the OOD test distribution.
 220 We repeat this construction for all OOD datasets and baselines. Each CIFAR InD dataset is split in
 221 half: the first 25,000 images train $\phi_{\mathcal{S}_{\text{in}}}$, and the remaining images contribute to $\mathcal{S}_{\text{wild}}$. For gradient
 222 computation, we use penultimate-layer weights because they have been shown to be informative for
 223 OOD detection (Huang et al., 2021a).

224 **Evaluation metrics.** We report three standard metrics that capture both detection and InD utility:
 225 (1) False Positive Rate (FPR95 $\downarrow$) of OOD samples when the True Positive Rate of InD samples is 95%,
 226 (2) Area Under the Receiver Operating Characteristic Curve (AUROC $\uparrow$), and (3) InD Classification
 227 Accuracy (InD Acc $\uparrow$).

Baselines. We organize baselines by the resource they use, which mirrors the contribution claim of Medix. For methods trained only on InD data ($\mathbb{P}_{\text{in}}$), we compare against Maximum Softmax Probability (MSP) (Hendrycks and Gimpel, 2016), ODIN (Liang et al., 2018), Mahalanobis Distance (Lee et al., 2018a), Energy Score (Liu et al., 2020b), ReAct (Sun et al., 2021), DICE (Sun and Li, 2022), KNN Distance (Sun et al., 2022), and ASH (Djurisic et al., 2022), all using a model trained with softmax cross-entropy loss. We also include contrastive-loss methods such as CSI (Tack et al., 2020a) and KNN+ (Sun et al., 2022). For methods that use both InD and wild data, we compare against Outlier Exposure (OE) (Hendrycks et al., 2019), energy-regularization learning (Liu et al., 2020b), and WOODS (Katz-Samuels et al., 2022a), which uses wild unlabeled data through constrained optimization. We further include recent baselines CONJ (Peng et al., 2024) and DRL (Zhang et al., 2024) for a broader comparison.

5.2 Experimental Setup

Following WOODS (Katz-Samuels et al., 2022a), we use a Wide ResNet (Zagoruyko, 2016) with 40 layers and width factor 2 for the InD classifier $\phi_{\mathcal{S}_{\text{in}}}$. We train it with stochastic gradient descent, momentum 0.9, weight decay 0.0005, and initial learning rate 0.1 for 100 epochs with cosine learning rate decay, batch size 128, and dropout 0.3. For the Medix filtering stage, ϵ and k are selected from $\{5e-5, 5e-4, 5e-3, 5e-2\}$ and $\{4k, 7k, 10k, 20k\}$, respectively, accounting for dataset size and optimizing OOD performance. For the OOD classifier g_{θ} , we initialize from the pre-trained InD classifier $\phi_{\mathcal{S}_{\text{in}}}$ and add a linear binary-classification layer on penultimate-layer features. The classifier uses learning rate 0.001 and is fine-tuned for 100 epochs according to Equation (5). We combine the binary classification loss with the InD classification loss, weighting the binary component by 10; all other training parameters match those used for $\phi_{\mathcal{S}_{\text{in}}}$.

Table 1: Main OOD detection comparison on CIFAR-10 as InD data, showing Medix against baselines trained with only $\mathbb{P}_{\text{in}}$ or with $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$. Performance is averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets										InD Acc $\uparrow$		
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES			Average	
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$		FPR95 $\downarrow$	AUROC $\uparrow$
Using $\mathbb{P}_{in}$ only													
MSP	48.49	91.89	59.48	88.20	30.80	95.65	52.15	91.37	59.28	88.50	50.04	91.12	94.84
ODIN	33.35	91.96	57.40	84.49	15.52	97.04	26.62	94.57	49.12	84.97	36.40	90.61	94.84
Mahalanobis	12.89	97.62	68.57	84.61	39.22	94.15	42.62	93.23	15.00	97.33	35.66	93.34	94.84
Energy	35.59	90.96	40.14	89.89	8.26	98.35	27.58	94.24	52.79	85.22	32.87	91.73	94.84
KNN	24.53	95.96	25.29	95.69	25.55	95.26	27.57	94.71	50.90	89.14	30.77	94.15	94.84
ReAct	40.76	89.57	41.44	90.44	14.38	97.21	33.63	93.58	53.63	86.59	36.77	94.48	94.84
DICE	35.44	89.65	46.83	86.69	6.32	98.68	28.93	93.56	53.62	82.20	34.23	90.16	94.84
ASH	6.51	98.65	48.45	88.34	0.90	99.73	4.96	98.92	24.34	95.09	17.03	96.15	94.84
CSI	17.30	97.40	34.95	93.64	1.95	99.55	12.15	98.01	20.45	95.93	17.36	96.91	94.17
KNN+	2.99	99.41	24.69	94.84	2.95	99.39	11.22	97.98	9.65	98.37	10.30	97.99	93.19
Using $\mathbb{P}_{in}$ and $\mathbb{P}_{wild}$													
OE	1.13	99.53	19.48	94.88	1.91	98.16	0.54	98.84	7.75	98.56	6.16	97.99	94.12
Energy (w/ OE)	5.24	98.72	14.66	96.18	2.35	99.30	4.85	98.62	10.51	97.10	7.52	97.98	94.24
WOODS	0.17	99.91	10.19	98.05	0.31	99.14	0.11	99.38	6.21	98.13	3.40	98.92	94.74
Medix	0.06	99.98	2.98	99.10	0.01	99.98	0.01	99.98	0.96	99.66	0.80	99.74	93.58
(Ours)	± 0.01	± 0.01	± 0.29	± 0.09	± 0.01	± 0.00	± 0.01	± 0.01	± 0.13	± 0.06	± 0.09	± 0.03	± 0.64

Table 2: Main OOD detection comparison on CIFAR-100 as InD data, evaluating whether Medix’s filtered wild-data supervision improves over baselines. Performance is averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets										InD ACC $\uparrow$		
	SVHN		PLACES365		LSUN-C		LSUN-RESIZE		TEXTURES			Average	
	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AUROC $\uparrow$			
	Using $\mathbb{P}_{\text{in}}$ only												
MSP	84.59	71.44	82.84	73.78	66.54	83.79	82.42	75.38	83.29	73.34	79.94	75.55	75.96
ODIN	84.66	67.26	87.88	71.63	55.55	87.73	71.96	81.82	79.27	73.45	75.86	76.38	75.96
Mahalanobis	57.52	86.01	88.83	67.87	91.18	69.69	21.23	96.00	39.39	90.57	59.63	82.03	75.96
Energy	85.82	73.99	80.56	75.44	35.32	93.53	79.47	79.23	79.41	76.28	72.12	79.69	75.96
KNN	66.38	83.76	79.17	71.91	70.96	83.71	77.83	78.85	88.00	67.19	76.47	77.08	75.96
ReAct	74.33	88.04	81.33	74.32	39.30	91.19	79.86	73.69	67.38	82.80	68.44	82.01	75.96
DICE	88.35	72.58	81.61	75.07	26.77	94.74	80.21	78.50	76.29	76.07	70.65	79.39	75.96
ASH	21.36	94.28	68.37	71.22	15.27	95.65	68.18	85.42	40.87	92.29	42.81	87.77	75.96
CSI	64.70	84.97	82.25	73.63	38.10	92.52	91.55	63.42	74.70	92.66	70.26	81.44	69.90
KNN+	32.21	93.74	68.30	75.31	40.37	86.13	44.86	88.88	46.26	87.40	46.40	86.29	73.78
	Using $\mathbb{P}_{\text{in}}$ and $\mathbb{P}_{\text{wild}}$												
OE	2.86	99.05	40.21	88.75	4.13	99.05	1.25	99.38	22.86	94.63	14.26	96.17	73.38
Energy (w/ OE)	2.71	99.34	34.82	90.05	3.27	99.18	2.54	99.23	30.16	94.76	14.70	96.51	72.76
WOODS	0.17	99.80	21.87	93.73	0.48	99.61	1.24	99.54	9.95	95.97	6.74	97.73	73.91
Medix	0.16	99.96	15.99	95.23	0.13	99.98	0.83	99.83	8.02	97.79	5.42	98.96	73.33
(Ours)	± 0.02	± 0.00	± 0.66	± 0.14	± 0.06	± 0.02	± 0.36	± 0.06	± 0.75	± 0.30	± 0.37	± 0.10	± 0.83

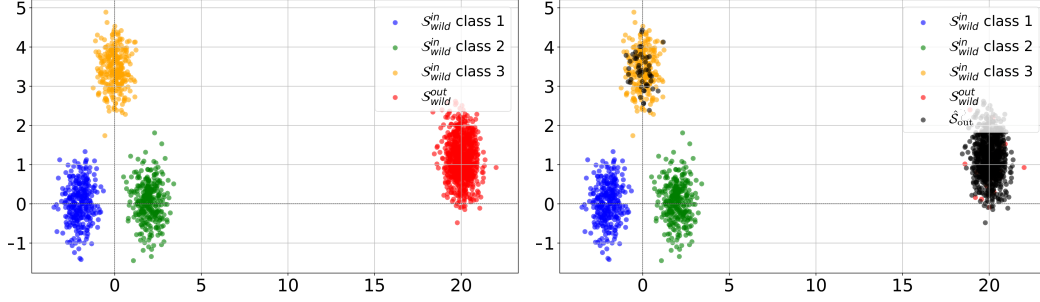


Figure 2: Representative visualization of the Medix filtering contribution on unlabeled wild data. (a) The wild set contains InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$ sampled from three multivariate Gaussian distributions and OOD data $\mathcal{S}_{\text{wild}}^{\text{out}}$. (b) Medix filters candidate outliers $\hat{\mathcal{S}}_{\text{out}}$ (black), with only 12.5% of $\hat{\mathcal{S}}_{\text{out}}$ coming from InD data $\mathcal{S}_{\text{wild}}^{\text{in}}$.

5.3 Results

The main results in Table 2 show that the Medix contribution translates into stronger OOD detection on CIFAR-100. Two observations are central. First, methods that use both InD and wild data generally outperform methods trained only on InD data, confirming the value of the resource targeted by Medix. Relative to KNN+, Medix reduces FPR95 by 52.31% on PLACES365 and 38.24% on TEXTURES. Second, Medix improves over methods that already use wild data: on CIFAR-100, it reaches an average FPR95 of 5.42%, a 1.32% improvement over WOODS, while maintaining InD accuracy of 73.33%. The small InD-accuracy difference is consistent with the setup: Medix trains on 25,000 labeled InD samples, whereas baselines that do not use wild data train on the full 50,000 CIFAR-100 images. Table 1 gives the corresponding CIFAR-10 results and shows the same contribution pattern. Medix outperforms all baselines and improves over WOODS by 7.21% on PLACES365 and 5.25% on TEXTURES in FPR95, indicating that the filtered OOD candidates provide useful supervision across InD datasets.

A representative visual example of Medix. We further isolate the C1 filtering stage by applying Algorithm 1 to a simple two-dimensional synthetic mixture. The purpose is interpretability rather than a new benchmark. InD data is sampled from three multivariate Gaussian distributions, corresponding to three classes, with mean vectors $[-2, 0]$, $[2, 0]$, and $[0, 2\sqrt{3}]$. Each class uses covariance $0.25 \cdot I$. OOD data is sampled from $\mathcal{N}([20, 2\sqrt{3}], 0.25 \cdot I)$. Figure 2 shows that most filtered points align with ground-truth outliers: Medix flags 87.5% of actual OOD samples as outliers, providing a direct visual check on the extraction mechanism.

Additional studies and insights. Appendix A extends the same contribution checks beyond the main tables. It includes hyperparameter sensitivity analysis (Appendix A.2), showing robustness to choices of ϵ and k ; an EWM-versus-geometric-median comparison (Appendix A.1), showing that EWM is more sensitive to distributional shifts and therefore better suited to our filter; a broader comparison with competitive contrastive-learning methods (Appendix A.3), where Medix remains strong; a large-scale unseen-OOD setting with $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$ (Appendix A.4); computation and memory profiling (Appendix A.6); pseudo-label quality analysis (Appendix A.5); and a comparison with semi-supervised open-set recognition methods (Appendix A.7).

6 Related Work

OOD detection has become a broad research area, and our contribution is best understood relative to three strands of prior work (Fort et al., 2021; Yang et al., 2024; Fang et al., 2022; Zhu et al., 2022; Yang et al., 2022; Wang et al., 2022c; Galil et al., 2023; Djuricic et al., 2023; Tao et al., 2023; Zheng et al., 2023; Wang et al., 2022b, 2023b; Uppaal et al., 2023; Zhu et al., 2023; Bai et al., 2023; Ming and Li, 2024; Zhang et al., 2023; Ghosal et al., 2024; Abbas et al., 2025). The first strand scores test inputs using a trained model or representation. Representative examples include distance-based methods (Lee et al., 2018b; Tack et al., 2020b; Ren et al., 2021; Schwag et al., 2021; Sun et al.,

286 2022; Du et al., 2022a; Ming et al., 2023; Ren et al., 2022), gradient-based scores (Huang et al.,
 287 2021b), energy-based scores (Liu et al., 2020c; Wang et al., 2021; Wu et al., 2023), confidence-based
 288 approaches (Bendale and Boulton, 2016; Hendrycks and Gimpel, 2017; Liang et al., 2018), and Bayesian
 289 methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Malinin
 290 and Gales, 2019; Wen et al., 2020; Kristiadi et al., 2020). Medix differs by using unlabeled wild data
 291 to construct candidate outliers before detector training.

292 The second strand improves OOD behavior through training-time regularization (Malinin and Gales,
 293 2018; Geifman and El-Yaniv, 2019; Hein et al., 2019; Meinke and Hein, 2020; Jeong and Kim, 2020;
 294 Liu et al., 2020a; Van Amersfoort et al., 2020; Yang et al., 2021; Wei et al., 2022; Du et al., 2022b,
 295 2023; Wang et al., 2023a). Such methods can reduce model confidence (Lee et al., 2017; Hendrycks
 296 et al., 2019) or increase energy (Liu et al., 2020c; Du et al., 2022c; Ming et al., 2022) on OOD data.
 297 The distinction is that many of these approaches assume access to auxiliary OOD samples, whereas
 298 Medix targets the harder case where only a mixed unlabeled wild set is available.

299 The third and most closely related strand uses unlabeled wild data (Zhou et al., 2021; Katz-Samuels
 300 et al., 2022b; He et al., 2023; Katz-Samuels et al., 2022a; Geng et al., 2025). Some of these works
 301 do not provide an explicit outlier-detection mechanism, while Du et al. (2024a,b) introduce filtering
 302 based on thresholding. Medix contributes a different filtering principle: a median-centric criterion
 303 that identifies outliers at the dataset level. This matters because Katz-Samuels et al. (2022a); Du et al.
 304 (2024a) assume batch-level mixing with a fixed InD/OOD ratio, whereas large outsourced datasets
 305 are typically mixed randomly without such structure. Positive-unlabeled learning also trains from
 306 positive and/or unlabeled data (Letouzey et al., 2000; Hsieh et al., 2015; Plessis et al., 2015; Niu
 307 et al., 2016; Gong et al., 2018; Chapel et al., 2020; Garg et al., 2021; Xu and Denil, 2021; Garg et al.,
 308 2022; Zhao et al., 2022; Acharya et al., 2022), but those methods focus on distinguishing $\mathbb{P}_{\text{out}}$ from
 309 $\mathbb{P}_{\text{in}}$ rather than simultaneously extracting outliers and training an OOD detector with provably low
 310 filtering error.

311 7 Conclusions

312 This paper contributes Medix, a median-centric framework for OOD detection with unlabeled in-
 313 the-wild data. The main methodological contribution is a greedy median-based filtering stage that
 314 extracts candidate outliers from a mixed wild set and uses them, together with labeled InD data, to
 315 train an OOD detector. The theoretical contribution is a two-sided analysis of this filter: we bound
 316 InD samples incorrectly flagged as outliers under contamination and concentration effects, and we
 317 bound OOD samples incorrectly retained under separation, concentration, and contamination effects.
 318 The empirical contribution is a broad validation across benchmarks and 20 baselines, where Medix
 319 reduces average FPR95 by 40.98% relative to strong baselines such as KNN+, outperforms methods
 320 such as WOODS and DRL, and achieves an outlier-extraction error rate as low as 12.5%. Together,
 321 these results support the paper’s central claim that robust median filtering can turn mixed unlabeled
 322 wild data into useful supervision for OOD detection.

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A Additional Studies

A.1 Element-wise Median vs Geometric Median

This study clarifies a design choice in the Medix filtering contribution: why use the element-wise median rather than the geometric median (Acharya et al., 2024) for outlier extraction. We test this choice with CIFAR-10 as the InD dataset and SVHN as the OOD dataset. The wild set $\mathcal{S}_{\text{wild}}$ contains 5k CIFAR-10 samples disjoint from the training set used to learn $\phi_{\mathcal{S}_{\text{in}}}$ and compute $\bar{\nabla}_{\text{in}}$. We then incrementally add SVHN OOD samples and track both the L_2 -norm deviation between $\bar{\nabla}_{\text{in}}$ and the wild-set median gradient and the proportion of OOD samples removed. Figure 3 shows that, for the same number of OOD samples in the wild set, the element-wise median identifies a substantially larger fraction of OOD samples than the geometric median. This supports the EWM design in Medix: its greater sensitivity to distributional shifts makes it a more effective filtering primitive for this method.

Table 3: Sensitivity of Medix’s filtering hyperparameters ϵ and k for OOD detection.

Method	FPR95↓	ϵ	k
DICE	88.35	—	—
ASH	21.36	—	—
CSI	64.70	—	—
KNN+	32.21	—	—
OE	2.86	—	—
Energy	2.71	—	—
Medix	0.16	0.005	20000
Medix	0.20	0.0005	20000
Medix	0.68	0.005	10000

A.2 Hyperparameter Selection and Sensitivity Analysis

This ablation evaluates whether Medix’s contribution depends on fragile choices of the filtering hyperparameters ϵ and k . We use CIFAR-100 as InD data and SVHN as OOD data. Table 3 shows that Medix remains robust across the tested values. With $\epsilon = 0.005$ and $k = 20000$, Medix achieves FPR95 of 0.16. Reducing ϵ to 0.0005 or halving k to 10000 keeps performance competitive, with FPR95 of 0.20 and 0.68, respectively. Thus, although tuning improves performance, Medix degrades gracefully and remains stronger than the baselines, which is important for deployment settings where exhaustive tuning may be impractical.

A.3 Comprehensive comparison with recent competitive methods

This comparison tests whether the empirical contribution persists against recent methods with strong and diverse training objectives. We include SSD+ (Sehwag et al., 2021), ProxyAnchor (Kim et al., 2020), and CIDER Ming et al. (2023), as well as CONJ (Peng et al., 2024), DRL (Zhang et al., 2024), Vim (Wang et al., 2022a), and VOS (Du et al., 2022c). Although several of these methods use contrastive objectives that Medix does not use, they provide a stronger comparison set. For this experiment, CIFAR-100 is the InD dataset and the ResNet-34 model¹ follows the setup in Ming et al. (2023); Zhang et al. (2024). The model is trained with stochastic gradient descent, momentum 0.9, weight decay 10^{-4} , 500 epochs, initial learning rate 0.5 with cosine scheduling, and batch size

¹Since ResNet-34 is the standard model across these works, we adopt the same model to ensure consistency and facilitate a fair comparison.

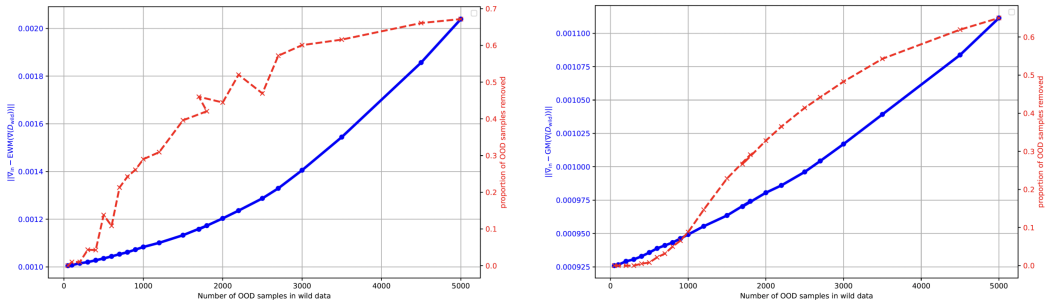


Figure 3: Median choice for the filtering contribution: element-wise median (EWM) identifies more OOD samples than geometric median (GM) in this wild-data study.

Table 4: Additional OOD detection comparison on CIFAR-100 against recent competitive baselines, including methods with contrastive objectives. Performance is averaged over five runs; best results are highlighted in **bold**.

Methods	OOD Datasets									
	SVHN		PLACES365		LSUN-C		TEXTURES		AVERAGE	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MSP	78.89	79.80	84.38	74.21	83.47	75.28	86.51	72.53	83.31	75.46
Mahalanobis	87.09	80.62	84.63	73.89	84.15	79.43	61.72	84.87	79.40	79.70
ODIN	70.16	84.88	82.16	75.19	76.36	80.10	85.28	75.23	78.49	78.85
Energy	66.91	85.25	81.41	76.37	59.77	86.69	79.01	79.96	71.78	82.07
ReAct	50.93	88.75	83.55	73.10	64.02	80.31	64.40	81.95	65.72	81.03
KNN	46.25	90.39	82.08	75.44	60.85	85.61	62.39	83.95	62.89	83.85
Vim	73.42	84.62	85.34	69.34	86.96	69.74	74.56	76.23	80.07	74.98
VOS	43.24	82.80	76.85	78.63	73.61	84.69	57.57	87.31	62.82	83.36
CSI	44.53	92.65	79.08	76.27	75.58	83.78	61.61	86.47	65.20	84.79
ProxyAnchor	87.21	82.43	70.10	79.84	37.19	91.68	65.64	84.99	65.04	84.74
SSD+	31.19	94.19	77.74	79.90	79.39	85.18	66.63	86.18	63.74	86.36
KNN+	39.23	92.78	80.74	77.58	48.99	89.30	57.15	88.35	56.53	87.00
ASH	52.96	90.19	72.62	76.38	75.16	76.52	56.17	86.75	64.23	82.46
CIDER	23.09	95.16	79.63	73.43	16.16	96.33	43.87	90.42	40.69	88.84
CONJ	46.19	90.44	80.81	75.83	60.45	85.90	62.13	83.77	62.40	83.99
DRL	20.15	94.07	76.64	77.55	16.97	94.63	31.97	92.09	36.43	89.59
Medix	0.48	99.87	24.52	93.42	0.73	99.84	8.99	97.92	8.68	97.76
(Ours)	± 0.05	± 0.02	± 1.76	± 0.36	± 0.08	± 0.03	± 0.72	± 0.24	± 0.65	± 0.16

512. We use the checkpoints provided by Ming et al. (2023)². Table 4 shows that Medix remains competitive against these baselines, including contrastive-learning methods, with an average FPR95 of 8.68% and average AUROC of 97.76%. This reinforces the empirical contribution while retaining the theoretical filtering guarantee that most alternatives do not provide.

A.4 Evaluation on Large-Scale, Complex Unseen OOD Settings

This study asks whether the Medix contribution extends to a large-scale unseen-OOD setting. The setting is harder because (1) the wild OOD data ($\mathbb{P}_{\text{out}}$) is large-scale, increasing data complexity and computational demand, and (2) the test-time OOD distribution ($\mathbb{P}_{\text{out}}^{\text{test}}$) differs from the wild OOD data used during training, i.e., $\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$, which better reflects challenging deployment scenarios.

We use CIFAR-100 as labeled InD data, 300K Random Images Hendrycks et al. (2019) as unlabeled wild OOD data, and SVHN as test OOD data. Because 300K Random Images and SVHN differ substantially, this setup evaluates whether filtered wild supervision can generalize beyond the wild OOD source. Table 5 compares Medix with other methods that also use wild data. Medix achieves lower FPR95 (41.29 ± 1.2) and higher AUROC (87.25 ± 0.9), indicating that the filtering stage remains useful even when

Table 5: Large-scale unseen-OOD evaluation, where wild OOD data and test OOD data differ.

Method	FPR ↓	AUROC ↑
OE	68.80 ± 2.8	82.89 ± 1.1
Energy (w/ OE)	69.81 ± 2.4	85.59 ± 1.0
WOODS	69.41 ± 2.7	86.76 ± 0.8
Medix (ours)	41.29 ± 1.2	87.25 ± 0.9

$\mathbb{P}_{\text{out}}^{\text{test}} \neq \mathbb{P}_{\text{out}}$.

A.5 Evaluating the Impact of Pseudo-Label Quality on OOD Filtering and Robustness

This study checks whether Medix’s filtering contribution is sensitive to pseudo-label quality, especially softmax confidence in $\hat{y}_{\tilde{x}_i}$. We evaluate whether a simple pre-filter that removes low-confidence pseudo-labels improves robustness. The experiment uses CIFAR-10 as InD data and LSUN-Resize as OOD data. We compute softmax probabilities for each pseudo-labeled sample and discard predictions below a confidence threshold of 0.6, removing 15.98% of samples from the training set.

The results show almost no performance difference between Medix with and without this confidence pre-filter. Both variants obtain FPR95 of 0.01%, and AUROC changes only from 99.98% to 99.99% with filtering. This indicates that the median-based extraction stage is already resilient to noisy or low-confidence pseudo-labels, and that this additional pre-filter is unlikely to provide meaningful gains.

²<https://github.com>

A.6 Evaluating Computation and Memory Efficiency in Medix Filtering

We next evaluate whether the Medix filtering stage is computationally feasible and memory-efficient. Profiling is performed on an NVIDIA A100-SXM4-80GB GPU with approximately 15,000 unlabeled samples for each InD–OOD pair, using CIFAR-10 and CIFAR-100 as InD datasets and LSUN-Resize as OOD data. Table 6 shows modest GPU memory usage and tractable wall-clock time, indicating that the filtering contribution can be applied to thousands of unlabeled samples without excessive memory overhead.

Table 6: Computation and memory profile of the Medix filtering stage on an NVIDIA A100 80GB GPU.

InD–OOD	Wall Clock Time (s)	Peak GPU Memory (MB)	Current GPU Memory (MB)
CIFAR10 – LSUN-Resize	4497.17	99.46	31.74
CIFAR100 – LSUN-Resize	5478.36	99.56	31.79

A.7 Comparison with Semi-Supervised Open-Set Recognition Methods

This comparison clarifies the boundary of the Medix contribution relative to semi-supervised open-set recognition methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024). In OpenML Saito et al. (2021), for example, the goal is to train a standard classifier in SSL while handling outliers. Our setting instead targets OOD detection: Medix extracts OOD samples from unlabeled wild data and trains a dedicated OOD detector when labeled OOD data is unavailable. Put differently, SSL methods (Saito et al., 2021; Fan et al., 2023; Hang and Zhang, 2024; Wang et al., 2024) primarily aim to improve classification despite outliers, whereas Medix is designed to improve OOD detection in an open-world setting without labeled OOD data.

A second distinction is how outliers are used. SSL methods generally treat outliers as noise to suppress during classifier training, whereas Medix treats them as informative candidates for training an OOD detector. This is why the filtering stage is central rather than incidental in our framework.

The techniques also differ. OpenMatch relies on consistency regularization across stochastic transformations, using soft constraints on classifier outputs to encourage smooth decision boundaries. Medix instead relies on median-based filtering, using the robustness of the median to identify potential OOD samples in a mixed wild set. The theoretical analysis then bounds filtering error through contamination and concentration effects.

B Broader Impacts and Limitations

The broader impact of this work is tied to its main contribution: improving OOD detection when only mixed unlabeled deployment data is available. More reliable detection can support safer model deployment in domains such as healthcare, autonomous systems, and finance, where unrecognized distribution shift can carry substantial risk. Future work could integrate Medix with generative models for outlier synthesis or adapt the filtering framework to dynamic environments where OOD distributions evolve over time.

A limitation is that Medix assumes access to an unlabeled mixture containing InD and OOD data, and real deployment mixtures may be noisy, corrupted, or inconsistently sampled. This assumption is not unique to Medix; it reflects the uncertainty of deployment data and is common in robust learning theory. We make the assumption explicit in the main theorems, state the geometric separation condition in Remark 4.3, and provide a looser guarantee in Theorem C.3 that removes sub-Gaussian tails. Thus the current contribution has simplifying assumptions, but they are interpretable and partially relaxable, leaving a clear path for future work on noisier wild-data settings.

688 C Proof of Theorem

689 To make the theoretical contribution easier to follow, Table 7 summarizes the notation used in the
 690 filtering bounds and proofs. The table covers gradient quantities, sample sizes, filtering sets, and
 691 concentration parameters. Unless stated otherwise, boldface denotes vectors, $\mathcal{X}$ denotes the input
 692 space, and gradients are computed with respect to a fixed pretrained model f_ϕ .

Table 7: Notation used in the Medix filtering guarantees and proofs. Bold symbols denote vectors and $\mathcal{X}$ denotes the input space.

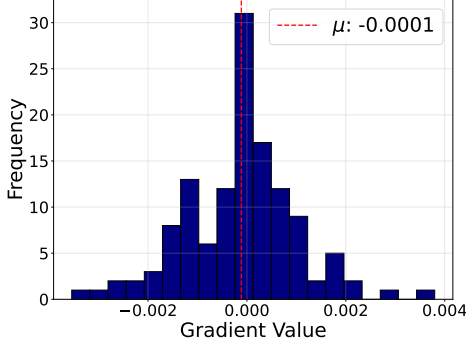
Symbol	Meaning
$\mathcal{X}$	Input space (e.g., images in $[0, 1]^d$)
f_ϕ	Model parameterized by ϕ
$\ell(f_\phi(x))$	Loss function evaluated at input x
$\nabla \ell(f_\phi(x))$	Gradient of loss at point x
$\mathbf{g}(x)$	Shorthand for $\nabla \ell(f_\phi(x))$, gradient vector
$\bar{\mathbf{g}}_{\text{in}}$	Empirical mean of gradients over InD points
m	Total number of unlabeled samples in $\mathcal{S}_{\text{wild}}$
$m_{\text{in}} = (1 - \pi)m$	Number of InD samples
$m_{\text{out}} = \pi m$	Number of OOD samples
$\pi \in (0, 1)$	OOD contamination ratio
$\mathcal{S}_{\text{wild}}$	Unlabeled mixture of InD and OOD data
$\mathcal{S}_{\text{in}}^*$	Inliers selected by the median filter
$\mathcal{S}_{\text{out}}^*$	OOD points selected by the median filter
$\mathcal{B}_\varepsilon(\bar{\mathbf{g}}_{\text{in}})$	Ball of radius ε centered at $\bar{\mathbf{g}}_{\text{in}}$
m_ε	Number of InD points inside $\mathcal{B}_\varepsilon$
m_ε^*	Number of selected inliers inside $\mathcal{B}_\varepsilon$
m_{sep}^*	Number of well-separated OODs in selected set
ERR_{in}	Fraction of InD points misclassified as OOD
ERR_{out}	Fraction of OOD points misclassified as InD
σ^2	Sub-Gaussian proxy variance of each gradient coordinate
ε	Threshold for gradient deviation, $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$
$\delta \in (0, 1)$	Confidence parameter for concentration bounds
d	Dimensionality of gradient vectors

693 C.1 Bound on the Inlier Misclassification Rate

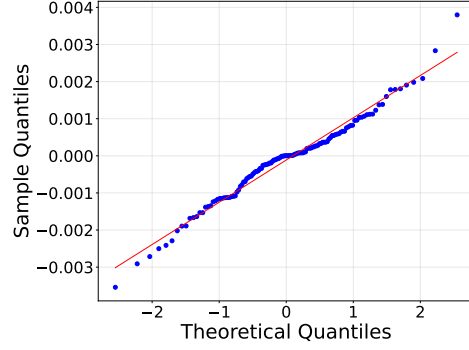
694 **Theorem C.1** (Inlier Misclassification Bound). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are*
 695 *i.i.d., and each coordinate is sub-Gaussian with variance proxy σ^2 . Let $\varepsilon = \sigma \sqrt{2 \log(2dm_{\text{in}})}$, and*
 696 *fix any confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification*
 697 *rate of the element-wise median (EWM) filtering rule satisfies:*

$$\text{ERR}_{\text{in}} \leq \underbrace{\frac{1}{m_{\text{in}}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}.$$

698 *Proof.* The proof mirrors the contribution claim for Medix: the median filter should not remove
 699 too many true InD samples. We proceed in two steps. In **Step 1**, we establish a high-probability
 700 concentration region around the InD mean gradient $\bar{\nabla}_{\text{in}}$ and call InD samples inside this region *good*
 701 *inliers*. In **Step 2**, we use a swapping argument for the EWM objective to show that excluding more
 702 than a $(2\eta + \frac{\pi}{2}(1 - \pi))$ fraction of good inliers, where η is the fraction of InD samples outside the
 703 concentration region, would contradict optimality. Combining these steps yields the stated upper
 704 bound on ERR_{in} .



(a) Histogram evidence for concentrated InD gradients.



(b) Q-Q evidence for light-tailed InD gradients.

Figure 4: Empirical support for the concentration assumption used in the Medix filtering proof. (left) Histogram of CIFAR-100 InD gradient values showing concentration around the mean with light tails, consistent with sub-Gaussianity. (right) Q-Q plot comparing empirical InD-gradient quantiles against a theoretical Gaussian distribution, showing alignment with the sub-Gaussian assumption.

705 **Step 1: Concentration of InD Gradients.** Let $\nabla \ell(f_\phi(\mathbf{x})) \in \mathbb{R}^d$ be the loss gradient (with
 706 respect to ϕ) for an InD sample $\mathbf{x}$. The concentration component assumes that each coordinate is
 707 sub-Gaussian with variance proxy σ^2 and common mean vector $\bar{\nabla}_{\text{in}}$. Thus, for each coordinate
 708 $j \in \{1, \dots, d\}$,

$$\mathbb{P}\left(|\nabla \ell(f_\phi(\mathbf{x}))_j - \bar{\nabla}_{\text{in},j}| \geq \epsilon\right) \leq 2 \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (6)$$

709 Define

$$\mathbf{v} := \nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}} \in \mathbb{R}^d.$$

710 The following norm relation converts coordinate-wise concentration into an ℓ_2 deviation bound:

$$\|\mathbf{v}\|_2 \leq \sqrt{d} \|\mathbf{v}\|_\infty,$$

711 Consequently, if $\|\mathbf{v}\|_2 > \epsilon \sqrt{d}$, then at least one coordinate j must satisfy $|v_j| > \epsilon$.

712 Combining (6) with a union bound over all d coordinates therefore yields

$$\mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (7)$$

713 Let $m_{\text{in}} = (1 - \pi)m$ be the number of InD points in $\mathcal{S}_{\text{wild}}$. For the filtering analysis, we call an InD
 714 point $\mathbf{x}$ *bad* if

$$\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}. \quad (8)$$

715 We next show that only a small fraction of the m_{in} inliers are bad with high probability.

716 Using the sub-Gaussian assumption and the union bound above, define the bad-inlier probability
 717 $p \in [0, 1]$ by

$$p := \mathbb{P}\left(\|\nabla \ell(f_\phi(\mathbf{x})) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}\right) \leq 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (9)$$

718 Here $\mathbf{x}$ denotes a generic InD sample, and the m_{in} inliers are assumed i.i.d.

719 Let X count bad inliers among the m_{in} samples. For each inlier $\mathbf{x}_i$, define

$$I_i := \begin{cases} 1, & \text{if } \|\nabla \ell(f_\phi(\mathbf{x}_i)) - \bar{\nabla}_{\text{in}}\|_2 > \epsilon \sqrt{d}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

720 Then X is the sum of these indicators:

$$X = \sum_{i=1}^{m_{\text{in}}} I_i. \quad (11)$$

721 Linearity of expectation gives

$$\mathbb{E}[X] = \sum_{i=1}^{m_{\text{in}}} \mathbb{E}[I_i] \quad (12)$$

$$= \sum_{i=1}^{m_{\text{in}}} \mathbb{P}(\mathbf{x}_i \text{ is bad}) \quad (13)$$

$$\stackrel{\text{from (9)}}{\leq} m_{\text{in}} \cdot p \quad (14)$$

$$\leq m_{\text{in}} \cdot 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \quad (15)$$

722 Since X is a sum of bounded Bernoulli variables, Hoeffding's inequality gives, for any $t > 0$,

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (16)$$

723 Setting

$$t := \sqrt{\frac{m_{\text{in}} \log(\frac{1}{\delta})}{2}}, \quad (17)$$

724 we obtain

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (18)$$

725 With probability at least $1 - \delta$, we therefore have

$$X \leq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (19)$$

726 Dividing both sides by m_{in} yields an upper bound on the fraction of bad inliers:

$$\frac{X}{m_{\text{in}}} \leq \frac{\mathbb{E}[X]}{m_{\text{in}}} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (20)$$

$$\stackrel{\text{from (12)}}{\leq} 2d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \quad (21)$$

$$= \underbrace{\frac{1}{m_{\text{in}}}}_{\text{Concentration term}} + 2\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} + \underbrace{\frac{\pi}{2(1-\pi)}}_{\text{Contamination term}}. \quad (22)$$

727 Let us denote this fraction by η . By setting $\epsilon = \sigma\sqrt{2\log(2dm_{\text{in}})}$ Thus, with probability at least
 728 $1 - \delta$, no more than an $O\left(d \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right) + \sqrt{\frac{\log(1/\delta)}{m_{\text{in}}}}\right)$ fraction of inliers have gradients lying
 729 outside the radius $\epsilon\sqrt{d}$ around $\bar{\nabla}_{\text{in}}$. These are precisely the *bad inliers*, while the remainder are
 730 called *good inliers*.

731 **Step 2: Contradiction via a Swapping Argument** The second step converts the concentration
 732 statement into a filtering guarantee. Suppose that $\mathcal{S}^*$ excludes too many InD points, specifically when

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (23)$$

733 then it is possible to construct a new set $\widehat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than that of $\mathcal{S}^*$,
 734 thereby contradicting the optimality of $\mathcal{S}^*$.

735 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 736 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (24)$$

737 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

738 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (25)$$

739 Define the swap size as

$$m'_\varepsilon := \min \{ m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}} \}. \quad (26)$$

740 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 741 available for swapping in.

742 Construct a new set $\widehat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 743 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\widehat{\mathcal{S}}| = |\mathcal{S}^*|.$$

744 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (27)$$

745 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (28)$$

746 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (29)$$

747 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (30)$$

748 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 749 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (31)$$

750 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (32)$$

751 Combining with the previous inequality gives:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (33)$$

752 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (34)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (35)$$

753 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (36)$$

754 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

755 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
756 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

757 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
758 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
759 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (37)$$

760 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (38)$$

761 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
762 increase it, so the median distance from μ_j does not increase and possibly decreases.

763 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (39)$$

764 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (40)$$

765 This strict improvement contradicts the assumption that $\mathcal{S}^*$ minimizes the median objective, complet-
766 ing the proof of the relaxed guarantee.

$$\Rightarrow \quad \text{ERR}_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (41)$$

767 □

768 C.2 Bound on the Outlier Retention Rate

769 We now establish a non-asymptotic upper bound on the fraction of OOD points that remain in
770 the subset returned by the MEDIX filter. This result mirrors the inlier misclassification bound
771 (Appendix C.1), but instead of relying on inlier concentration, it exploits a separation assumption
772 between the OOD and InD gradient means.

773 **Theorem C.2** (Outlier Retention Bound under Vector Separation). *Assume that for every OOD point*
774 *$x \sim P_{\text{out}}$, the gradient $g(x) := \nabla \ell(f_{\varphi_{\text{in}}}(x)) \in \mathbb{R}^d$ satisfies:*

775 **(I) Sub-Gaussianity.** *Each coordinate $g_j(x)$ is sub-Gaussian with variance proxy σ_{out}^2 .*

776 **(II) Vector Mean Separation.** *The mean OOD gradient vector μ_{out} satisfies*

$$\|\mu_{\text{out}} - \bar{\nabla}_{\text{in}}\|_2 \geq \Delta \sqrt{d}, \quad \text{for some } \Delta > 0.$$

777 Fix any tolerance $\varepsilon \in (0, \Delta)$ and confidence parameter $\delta \in (0, 1)$. Define:

$$p_{\text{out}}(\varepsilon) := 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right), \quad \eta_{\text{out}} := p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}.$$

778 Then with probability at least $1 - \delta$, the outlier retention rate satisfies:

$$\text{ERR}_{\text{out}} := \frac{|\mathcal{S}_{\text{in}}^* \cap \text{OOD}|}{m_{\text{out}}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}. \quad (42)$$

779 **Proof. Step 1: Concentration of OOD Gradients.**

780 By assumption, each coordinate of $g(x)$ is sub-Gaussian, and the mean vector is separated from $\bar{\nabla}_{\text{in}}$
781 by at least $\Delta\sqrt{d}$. Therefore, for any OOD point x , the event

$$\|g(x) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \quad (43)$$

782 has probability at most $p_{\text{out}}(\varepsilon)$.

783 **Counting the “ambiguous” OOD points.** For every OOD example x_i define the indicator

$$J_i := \mathbf{1}\left\{\|g(x_i) - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d}\right\}, \quad i = 1, \dots, m_{\text{out}}.$$

784 Each $J_i \sim \text{Bernoulli}(p_{\text{out}}(\varepsilon))$ and the J_i ’s are independent. Let

$$Y := \sum_{i=1}^{m_{\text{out}}} J_i, \quad \mu := \mathbb{E}[Y] = m_{\text{out}} p_{\text{out}}(\varepsilon).$$

785 Because the indicators J_i take values in $\{0, 1\}$, Hoeffding’s inequality states that for any $s > 0$

$$\Pr\{Y - \mu \geq s\} \leq \exp\left(-\frac{2s^2}{m_{\text{out}}}\right).$$

786 Choose $s = \sqrt{\frac{m_{\text{out}} \log(1/\delta)}{2}}$; then (C.2) equals δ , and with probability at least $1 - \delta$

$$Y \leq m_{\text{out}} \left(p_{\text{out}}(\varepsilon) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}}\right) = m_{\text{out}} \eta_{\text{out}}.$$

787 Define the high-probability event

$$\mathcal{E}_1 := \{Y \leq m_{\text{out}} \eta_{\text{out}}\}, \quad (35)$$

788 which holds with probability at least $1 - \delta$.

789 **Step 2: Contradiction via a Swapping Argument** Suppose, toward contradiction, that

$$\text{ERR}_{\text{out}} > \eta_{\text{out}} + \frac{1 - \pi}{2\pi}. \quad (44)$$

790 Let $m_{\text{out}}^* := |\mathcal{S}_{\text{in}}^* \cap \text{OOD}|$, and define $m_\varepsilon^* := \#\{\text{ambiguous OODs in } \mathcal{S}_{\text{in}}^*\}$. Then, under $\mathcal{E}_1$,

$$m_{\text{sep}}^* := m_{\text{out}}^* - m_\varepsilon^* > \frac{1 - \pi}{2\pi} m_{\text{out}} = \frac{1}{2} m_{\text{in}}. \quad (45)$$

791 This inequality creates the contradiction used in the outlier-retention contribution: more than half the
792 points in $\mathcal{S}_{\text{in}}^*$ would be separated OODs with gradients outside the ball

$$B_\varepsilon := \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (46)$$

By the definition of the coordinate-wise median, this implies that in every coordinate, the majority of entries in $G_{\mathcal{S}_{\text{in}}}^{(j)}$ lie outside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, hence:

$$\|\text{EWM}(G_{\mathcal{S}_{\text{in}}}^*) - \bar{\nabla}_{\text{in}}\|_2^2 \geq \varepsilon^2 d. \quad (47)$$

Now consider a new subset $\hat{\mathcal{S}}$ formed by swapping the m_{sep}^* separated OODs in $\mathcal{S}_{\text{in}}^*$ with an equal number of InD points whose gradients lie inside B_ε . Such inliers exist because fewer than $m_{\text{out}}\eta_{\text{out}}$ are ambiguous, and the total number of InD points is m_{in} .

Then at least half the entries in every coordinate of $G_{\hat{\mathcal{S}}}^{(j)}$ lie inside the interval $[\bar{\nabla}_{\text{in},j} - \varepsilon, \bar{\nabla}_{\text{in},j} + \varepsilon]$, implying:

$$\|\text{EWM}(G_{\hat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2^2 \leq \varepsilon^2 d. \quad (48)$$

Combining with (47) yields a contradiction:

$$\|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\hat{\mathcal{S}}})\|_2^2 < \|\bar{\nabla}_{\text{in}} - \text{EWM}(G_{\mathcal{S}_{\text{in}}}^*)\|_2^2. \quad (49)$$

This strict improvement contradicts the optimality of $\mathcal{S}_{\text{in}}^*$. Hence the assumed excessive outlier retention cannot occur, and with probability at least $1 - \delta$, the outlier retention rate is bounded by

$$\text{ERR}_{\text{out}} \leq 2d \exp\left(-\frac{(\Delta - \varepsilon)^2}{2\sigma_{\text{out}}^2}\right) + \sqrt{\frac{\log(1/\delta)}{2m_{\text{out}}}} + \frac{1 - \pi}{2\pi}, \quad (50)$$

□

C.3 Theorem C.3 without Sub-Gaussianity

Theorem C.3 (Inlier Misclassification Bound without Sub-Gaussianity). *Assume that the gradients of InD points in $\mathcal{S}_{\text{wild}}$ are i.i.d., and that each coordinate has variance σ^2 and finite fourth moment bounded by μ_4 , i.e.,*

$$\mathbb{E}[(\nabla \ell(f_\phi(x))_j - \bar{\nabla}_{\text{in},j})^2] = \sigma^2, \quad \mathbb{E}[(\nabla \ell(f_\phi(x))_j - \bar{\nabla}_{\text{in},j})^4] \leq \mu_4.$$

Let $m_{\text{in}} = (1 - \pi)m$ denote the number of InD samples in $\mathcal{S}_{\text{wild}}$, and fix any tolerance $\varepsilon > \sigma$ and confidence level $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, the inlier misclassification rate of the element-wise median (EWM) filtering rule satisfies:

$$\text{ERR}_{\text{in}} \leq \underbrace{2 \left(\frac{\mu_4 - \sigma^4}{d(\varepsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}} \right)}_{\text{Concentration term}} + \underbrace{\frac{\pi}{2(1 - \pi)}}_{\text{Contamination term}}.$$

Differences with the original bound. This appendix contribution clarifies what is lost when the sub-Gaussian assumption is removed. The original bound features the decay term

$$2 \exp\left(-\frac{\varepsilon^2 d}{2\sigma_{\text{sub}}^2}\right)$$

813 which decreases exponentially with dimension d and is therefore tight in high dimensions. The
 814 finite-fourth-moment version replaces it with

$$\frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}$$

815 which decays as $O(1/d)$ and is therefore looser. The trade-off is interpretability and weaker assump-
 816 tions: the original guarantee requires sub-Gaussian tails, whereas the new one allows heavier tails
 817 through finite fourth moments. Both bounds retain the finite-sample term

$$\sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}},$$

818 which shrinks as m_{in} increases, independent of the tail assumption.

819 *Proof.* The proof preserves the same contribution structure as the sub-Gaussian theorem while
 820 weakening the tail assumption. It has two steps: (1) establish a concentration bound on the fraction
 821 of “bad” InD samples, i.e., samples far from ∇_{in} , without using sub-Gaussianity; and (2) reuse the
 822 swapping argument to bound the inlier misclassification rate. The only substantive change is that the
 823 sub-Gaussian tail bound is replaced by a moment-based bound derived from the finite fourth moment
 824 assumption.

825 **Step 1: Concentration of InD Gradients** Define the gradient deviation for an InD sample x by

$$v = \nabla \ell(f_\phi(x)) - \nabla_{\text{in}} \in \mathbb{R}^d, \quad (51)$$

826 where $\nabla_{\text{in}} = \frac{1}{n} \sum_{(x_i, y_i) \in S_{\text{in}}} \nabla \ell(f_\phi(x_i))$ is the mean gradient over the labeled InD training set S_{in} ,
 827 and we assume ∇_{in} is the population mean. An InD sample is “bad” if

$$\|v\|_2 > \epsilon \sqrt{d}, \quad (52)$$

828 and “good” otherwise. The goal is to bound the bad-inlier fraction among the m_{in} InD samples in
 829 S_{wild} .

830 Since $\|v\|_2^2 = \sum_{j=1}^d v_j^2$, we control the ℓ_2 norm through squared coordinate deviations. Assume that
 831 the coordinates $v_j = \nabla \ell(f_\phi(x))_j - \nabla_{\text{in},j}$ are independent across $j = 1, \dots, d$ and satisfy

$$\mathbb{E}[v_j] = 0 \quad (\text{assuming } \nabla_{\text{in},j} = \mathbb{E}[\nabla \ell(f_\phi(x))_j]), \quad (53)$$

$$\mathbb{E}[v_j^2] = \sigma^2, \quad \mathbb{E}[v_j^4] \leq \mu_4. \quad (54)$$

832 Thus

$$\mathbb{E}[\|v\|_2^2] = \mathbb{E}\left[\sum_{j=1}^d v_j^2\right] = d\sigma^2, \quad (55)$$

$$\text{Var}(\|v\|_2^2) = \text{Var}\left(\sum_{j=1}^d v_j^2\right) = \sum_{j=1}^d \text{Var}(v_j^2). \quad (56)$$

833 For each coordinate, the variance of v_j^2 is bounded by

$$\text{Var}(v_j^2) = \mathbb{E}[v_j^4] - (\mathbb{E}[v_j^2])^2 \leq \mu_4 - \sigma^4, \quad (57)$$

834 so:

$$\text{Var}(\|v\|_2^2) \leq d(\mu_4 - \sigma^4). \quad (58)$$

835 Using Chebyshev's inequality for $\|v\|_2^2$:

$$\mathbb{P}(|\|v\|_2^2 - d\sigma^2| \geq t) \leq \frac{d(\mu_4 - \sigma^4)}{t^2}. \quad (59)$$

836 We need $\mathbb{P}(\|v\|_2 > \epsilon\sqrt{d})$, which corresponds to $\|v\|_2^2 > \epsilon^2 d$. Set the threshold:

$$\|v\|_2^2 > \epsilon^2 d \implies \|v\|_2^2 - d\sigma^2 > \epsilon^2 d - d\sigma^2 = d(\epsilon^2 - \sigma^2). \quad (60)$$

837 Thus:

$$\mathbb{P}(\|v\|_2^2 > \epsilon^2 d) = \mathbb{P}(\|v\|_2^2 - d\sigma^2 > d(\epsilon^2 - \sigma^2)) \leq \frac{d(\mu_4 - \sigma^4)}{[d(\epsilon^2 - \sigma^2)]^2} = \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}, \quad (61)$$

838 provided $\epsilon > \sigma$ so that $\epsilon^2 - \sigma^2 > 0$. Define:

$$p = \mathbb{P}(\|\nabla \ell(f_\phi(x)) - \nabla_{\text{in}}\|_2 > \epsilon\sqrt{d}) \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (62)$$

839 Let X be the number of bad inliers among the m_{in} InD samples in S_{wild} , where each InD sample is
840 bad with probability at most p . Then

$$\mathbb{E}[X] \leq m_{\text{in}}p \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2}. \quad (63)$$

841 Because $X = \sum_{i=1}^{m_{\text{in}}} I_i$, where $I_i = 1$ when the i -th InD sample is bad and 0 otherwise, the bounded
842 indicators allow Hoeffding's inequality:

$$\mathbb{P}(X \geq \mathbb{E}[X] + t) \leq \exp\left(-\frac{2t^2}{m_{\text{in}}}\right). \quad (64)$$

843 Set $t = \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}$, so:

$$\mathbb{P}\left(X \geq \mathbb{E}[X] + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}\right) \leq \delta. \quad (65)$$

844 With probability at least $1 - \delta$:

$$X \leq m_{\text{in}} \cdot \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{m_{\text{in}} \log(1/\delta)}{2}}. \quad (66)$$

845 Therefore the bad-inlier fraction satisfies

$$\eta = \frac{X}{m_{\text{in}}} \leq \frac{\mu_4 - \sigma^4}{d(\epsilon^2 - \sigma^2)^2} + \sqrt{\frac{\log(1/\delta)}{2m_{\text{in}}}}. \quad (67)$$

846 This η upper-bounds the fraction of InD samples whose gradients deviate from ∇_{in} by more than
847 $\epsilon\sqrt{d}$ using only finite fourth moments and coordinate independence.

848 **Step 2: Contradiction via a Swapping Argument** The second step reuses the Medix optimality
 849 argument. If $\mathcal{S}^*$ excludes too many InD samples, namely if

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (68)$$

850 then one can construct a new set $\hat{\mathcal{S}}$ whose EWM is strictly closer to $\bar{\nabla}_{\text{in}}$ than the EWM of $\mathcal{S}^*$,
 851 contradicting the optimality of $\mathcal{S}^*$.

852 Let m_ε be the number of indices i such that either x_i is OOD or x_i is an InD point whose gradient
 853 lies outside the closed ℓ_2 ball

$$B_\varepsilon = \left\{ g \in \mathbb{R}^d : \|g - \bar{\nabla}_{\text{in}}\|_2 \leq \varepsilon\sqrt{d} \right\}. \quad (69)$$

854 Let m_{in}^* be the number of InD points contained in the candidate minimizer $\mathcal{S}^*$.

855 Assume for contradiction that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (70)$$

856 Define the swap size as

$$m'_\varepsilon := \min \{m_\varepsilon, (\text{ERR}_{\text{in}} - \eta)m_{\text{in}}\}. \quad (71)$$

857 This quantity counts how many good inliers (those inside B_ε) are currently excluded from $\mathcal{S}^*$ and
 858 available for swapping in.

859 Construct a new set $\hat{\mathcal{S}}$ by removing m'_ε points from $\mathcal{S}^*$ that lie outside B_ε (OOD points or bad inliers)
 860 and inserting m'_ε good inliers from B_ε in their place. This maintains cardinality:

$$|\hat{\mathcal{S}}| = |\mathcal{S}^*|.$$

861 Recall that

$$\text{ERR}_{\text{in}} := \frac{m_\varepsilon - m_{\text{in}}^*}{m_{\text{in}}}, \quad (72)$$

862 which measures the fraction of good inliers excluded from $\mathcal{S}^*$. Rearranging, we obtain:

$$m_\varepsilon - m_{\text{in}}^* = \text{ERR}_{\text{in}} \cdot m_{\text{in}}. \quad (73)$$

863 Now we know, for contradiction, that

$$\text{ERR}_{\text{in}} > 2\eta + \frac{\pi}{2(1-\pi)}, \quad (74)$$

864 Subtracting η from both sides and multiplying by m_{in} , we obtain:

$$(\text{ERR}_{\text{in}} - \eta)m_{\text{in}} > \left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}}. \quad (75)$$

865 We know that at most a η fraction of inliers are outside B_ε , and the fraction of OOD points among all
 866 samples is π . Therefore, the total number of samples in the dataset that lie outside B_ε satisfies:

$$m_\varepsilon \leq \eta m_{\text{in}} + \frac{\pi}{1-\pi} m_{\text{in}} = \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}}. \quad (76)$$

867 It follows that

$$\left(\eta + \frac{\pi}{2(1-\pi)} \right) m_{\text{in}} > \frac{1}{2} \cdot \left(\eta + \frac{\pi}{1-\pi} \right) m_{\text{in}} \geq \frac{m_\varepsilon}{2}. \quad (77)$$

868 Combining with the previous inequality gives:

$$(ERR_{\text{in}} - \eta)m_{\text{in}} > \frac{m_\varepsilon}{2}, \quad (78)$$

869 So $m'_\varepsilon = m_\varepsilon/2 + \eta m_{\text{in}}$, which implies

$$m_\varepsilon - m'_\varepsilon = \frac{m_\varepsilon}{2} - \eta m_{\text{in}}, \quad (79)$$

$$m_{\text{in}}^* + m'_\varepsilon = m_{\text{in}}^* + \frac{m_\varepsilon}{2} + \eta m_{\text{in}}. \quad (80)$$

870 Because $\eta m_{\text{in}} > 0$, it follows that

$$m_{\text{in}}^* + m'_\varepsilon > m_\varepsilon - m'_\varepsilon, \quad (81)$$

871 so the points lying inside B_ε form a strict majority in $\widehat{\mathcal{S}}$.

872 Fix any coordinate $j \in \{1, \dots, d\}$. Let $X^{(j)} = \{x_1^{(j)}, \dots, x_m^{(j)}\}$ be the j -th coordinate values of all
873 gradients in $\mathcal{S}^*$ and $\widehat{X}^{(j)} = \{\widehat{x}_1^{(j)}, \dots, \widehat{x}_m^{(j)}\}$ the same in $\widehat{\mathcal{S}}$. Let $\mu_j = \bar{\nabla}_{\text{in},j}$.

874 Each inserted gradient lies inside the interval $I_j = [\mu_j - \varepsilon, \mu_j + \varepsilon]$, and each removed gradient lies
875 outside I_j . So for some coordinate j^* , the number of values inside I_{j^*} increases from below $m/2$ to
876 above $m/2$ due to the swaps. Therefore:

$$\text{med}(X^{(j^*)}) \notin I_{j^*}, \quad \text{med}(\widehat{X}^{(j^*)}) \in I_{j^*}, \quad (82)$$

877 which implies

$$|\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| \leq \varepsilon \quad \text{and} \quad |\text{med}(\widehat{X}^{(j^*)}) - \mu_{j^*}| < |\text{med}(X^{(j^*)}) - \mu_{j^*}|. \quad (83)$$

878 For the remaining coordinates $j \neq j^*$, the swapped points either leave the majority unchanged or
879 increase it, so the median distance from μ_j does not increase and possibly decreases.

880 Thus, component-wise:

$$|\text{EWM}_j(G_{\widehat{\mathcal{S}}}) - \mu_j| \leq |\text{EWM}_j(G_{\mathcal{S}^*}) - \mu_j|, \quad (84)$$

881 with strict inequality in at least one coordinate j^* , implying

$$\|\text{EWM}(G_{\widehat{\mathcal{S}}}) - \bar{\nabla}_{\text{in}}\|_2 < \|\text{EWM}(G_{\mathcal{S}^*}) - \bar{\nabla}_{\text{in}}\|_2. \quad (85)$$

882 This strict improvement contradicts the assumption that $\mathcal{S}^*$ minimizes the median objective, complet-
883 ing the proof of the relaxed guarantee.

$$\Rightarrow \quad ERR_{\text{in}} \leq 2\eta + \frac{\pi}{2(1-\pi)}. \quad (86)$$

884 □